Investigating Semantic Similarity's Role in Word Analogies

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Abstract

Inspired by the parallelogram model for word analogy, given proportional word analogies such as King: Queen:: Man: Woman, their offset similarity is viewed as relational similarity between the words $\mathbf{king} - \mathbf{queen} = \mathbf{man} - \mathbf{woman} = \mathbf{r}$. To that end, despite numerous studies, the role of semantic similarity in offset or relational similarity remains unclear. As such, various works framing the problem as that of analogy completion $x : x^* :: y : \underline{?}$, show that often finding the answer y^*, by choosing the word most similar to y, gives better results than finding it with the offset vector $y^* = y + (x^* - x)$. In this paper, we view word analogy as that of relational mapping. Accordingly, given a query $x - x^* = \underline{?}$, the task is to find word pairs that are related to one another as words in the query are related. Accordingly, noting that word vectors are obtained through word co-occurrence, and distinguishing between similarity due to relatedness (such as similarity between coffee and cup) and semantic similarity (such as similarity between coffee and tea), we show that in common word embedding models, word vectors exhibit similarities with both related words, and also semantically similar words. Additionally, we exactly show the configurations where word similarity can limit offset similarity's success in distinguishing between relation similarity and mere similarity between words.

1 Introduction

Analogy and analogical reasoning are a crucial aspect of human cognition. As such, understanding them is of significant importance. One common example of analogy is the so-called proportional word analogies, traditionally used to evaluate word embedding models. To that end, given that word embedding models such as fasttext and word2vec obtain the representation of a given word using words that accompany it in textual contexts, words

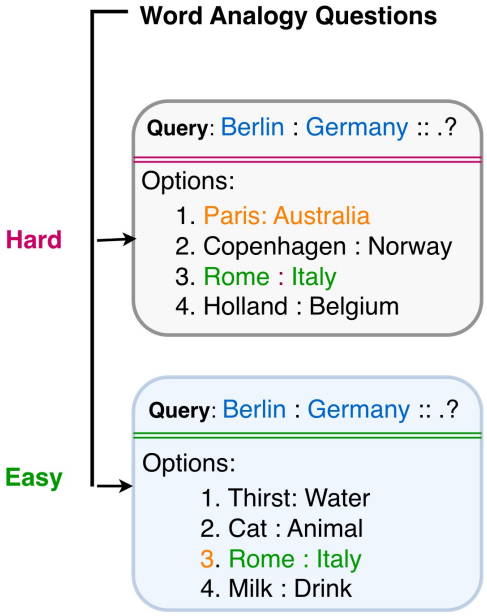


Figure 1: Word Similarity, as the difficulty factor for word analogy questions. Distractor with similarity to the query is orange. True answer is green.

can be similar in two main possible ways: Firstly, two given words that often appear together in textual contexts such as coffee and cup, may have similar embeddings. Assuming words appearing together in textual contexts are related, we can call such similarity in word vectors as similarity due to relatedness. It is to note that, in literature, such similarity is also called topical similarity, or domain similarity (Hill et al., 2015). Secondly, words that do not appear in the same contexts can still have similarity in their vectors, given that there are words that often accompany both words. Arguing that the common words in different contexts represent relation similarity involving our given words, we can call such similarity, as similarity due to relation similarity. Considering the set theoretic view of similarity, where semantic similarity is described as feature-mapping process (Tversky, 1977), the view of semantic similarity being due to relation

similarity is coherent with literature. For example, consider proportional analogy word:language :: note : music; one can imagine there are words that express the relations between words (word, language), similar to the relations between words (note , music) in various contexts. Noting the one-to-one correspondence between head (word, language) and tail (note , music) words, there is pairwise similarity between head and tail words. The intuition for such similarity in similarly related words (word pairs that are similarly related) is that relations have directions, therefore we expect "word" to be similar to "note", while "language" should be similar to "music". Semantic similarity from the Google word analogy dataset is illustrated in Figure 2.

Word embedding models are shown to capture word relations without being explicitly trained on them. As such, inspired by the parallelogram model for analogy, the similarity between offset of word pairs is viewed as the relation similarity between word pairs $\mathbf{king} - \mathbf{queen} = \mathbf{man} - \mathbf{woman} = \mathbf{r}$, with $\mathbf{r}$ being referred to as the offset vector. To that end, the similarity between offsets of such words, and the similarity between words themselves have been a topic of research. Accordingly, given proportional word analogies $x : x^* :: y : y^*$, studies framing the problem as multi-choice analogy completion problem $x : x^* :: y : \underline{?}$, show that for offset vector to be helpful in answering the question, y and y^* should be very close to each other (Similar). Additionally, studies also show that similarity alone is sometimes more successful at the analogy completion problem. In this paper, we frame the problem as that of relational mapping (Gentner and Forbus, 2011) $x : x^* :: \underline{?}$. This simple change of problem representation makes the word analogy problem consistent with the structural mapping definition of analogy (Gentner, 1983), where objects, as well as relations are mapped. Accordingly, we show that for offset of analogous words to be similar, important is semantic similarity between head (x, y) and tail (x^*, y^*) terms. Moreover, the linear connection between pairwise head and tail similarity and that of offset similarity is also consistent with the notion of re-representation from cognitive science. As such, according to re-representation, during analogical reasoning humans change the representation of entities or objects so that the relation matches (Gentner and Forbus, 2011).

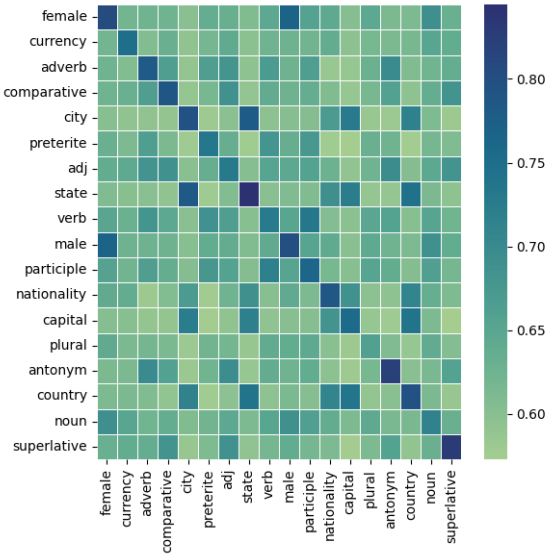


Figure 2: Similarity of relationally similar words using the word analogy dataset Google, wherein each head and tail word for a given relation r corresponds to a unique concept (for the relation "citizen of", the head word is a person, and the tail word is nationality).

2 Related Works

The related work of this paper concerns 2 categories of research papers: those that investigate the emergence of linear analogies in word embedding models, and those that explore the relation between word similarity and offset similarity.

Vector Arithmetic Based Approaches for Word Analogy: Pennington et al. (2014) show that for the given proportional analogies, offset vectors will be similar given that the proportion $p(w|x)/p(w|x^*) = p(w|y)/p(w|y^*)$ holds for all words w in the vocab. Furthermore, Ethayarajh et al. (2019) prove the above conjecture theoretically. Moreover, using a word co-occurrence matrix, Korchinski et al. (2025); Karkada et al. (2025) show how simple linear analogies in the word embedding space emerge.

Word Similarity and Offset Similarity in Literature: Various works study such a connection in the context of analogy completion in a multiple-choice question-answering format, such as King : Queen :: Man :? , where it is commonly solved by finding the word closest to man+(queen-king). As such, the parallelogram revisited (Peterson et al., 2020) shows that word similarity alone often gives better results on the task than offset vectors. Additionally, Linzen (2016) argue that, given the multiple-choice analogy completion question

$x : x^* :: y :?$, when x, x^* and y are added to the distractors, the performance drops significantly. He also shows that, for analogy completion, the change in performance when offsets are not used is not much different from that when offsets are used. Rogers et al. (2017) further extend the work of Linzen (2016) and show that as the similarity between y and y^* decreases, so does the success of the offset at analogy completion. In other words, Rogers et al. (2017); Linzen (2016) point out that in order for the offset vector to work, y should be very similar to y^*, and the help due to the use of the offset vector is insignificant, whereas in this work we show that y and y^* may be similar as they are related and appear often in same contexts; which is different from desired semantic similarity between words. To that, the similarity between y and any other word x_i is desirable, when it involves some forms of relation similarity indicated by some proportional analogies: $x_i : x_i^* :: y : y_j^*$, for some i and j.

3 Common Word Analogy Benchmarks

Ushio et al. (2021) categorizes word analogy datasets into psychometric analogy tests and lexical semantics benchmarks. For lexical semantics benchmarks, there are Google (Mikolov et al., 2013b) and BATS (Gladkova et al., 2016). Google contains only semantic and morphological relations, whereas BATS includes a more diverse set of analogies categorized into lexicographic, encyclopedic, derivational, and inflectional morphology (Ushio et al., 2021). From Psychometric Analogy Tests we have SAT (Turney et al., 2005), U4 (Ushio et al., 2021) and U2 (Boteanu and Chernova, 2015). SAT is a college admissions test designed to be challenging for humans. U2 was collected from the English learning website by (Boteanu and Chernova, 2015). U2 is intended for school students of grades 2-12. Similarly, U4 is collected by (Ushio et al., 2021) from the same website and is categorized by difficulty level. Furthermore, EKar (Chen et al., 2022) is constructed from the Civil Service Exams in China and is quite challenging for language models to solve. Various datasets used in this paper are shown in the Table 1.

4 Motivation: Word Similarity and Word Analogy

Noting types of similarity in word vectors, namely: **Similarity Due to Relatedness**, which can be attributed to the frequent appearance of words in the same textual contexts, and **Similarity due to Relation Similarity**, which can be attributed to relation similarity between entities or words expressed in various textual contexts; and the notion of re-representation from cognitive science, we argue that studying semantic similarity and relation similarity should inlvove framing the problem as that of relation mapping rather than analogy completion, as firstly: the analogy completion is influenced by similarity due to relatedness, and secondly, given that word analogies are a form of analogy, relational mapping format of word analogy as was introduced, better match the common accepted understanding of analogies. To that end, we further hypothesize that: in human constructed word analogy benchmarks, when forming the multiple choice word analogy questions, preference is given for distractors with some form of similarity with the query. For example, in Figure 3, given the query words (word, language) and the distractor words (week, year), "word" and "week" are units of measurement in their respective contexts, text, and time. As such, if the word embedding model is trained on texts that express such properties of the words "word" and "week," respectively, their vectors may be, to some extent, similar. Accordingly,

Dataset	choices	(valid/test)
Ekar	4	(262/870)
Scan	3-153	178/1616
Google	4	(50/500)
BATs	4	(199/1799)
U4	5/4/3	(48/432)
U2	5/4/3	(24/228)
SAT	5	(48/432)

Table 1: Word Analogy Datasets used in this work.

QUERY: word:language
d_1: paint:portrait
d_2: poetry:rhythm
d_3: **note:music** ← *answer*
d_4: tale:story
d_5: week:year

Figure 3: A typical word analogy question. Note the query, the answer(d_3) and distractors (d_i, with $i \neq 3$). The types of similarity between query and distractors: semantic similarity (color) , and similarity due to relatedness (color).

given our assumption regarding types of similarity, and our motivation regarding the connection between offset similarity and semantic similarity, we show evidence indicating that in word embedding models, word vectors for relationally similar words might be similar as a result of semantic similarity, and word vectors for related words too might be similar as they often appear in same textual contexts. Additionally, word similarity without relational similarity (such as Berlin-Germany!=Paris-Denmark) is one the limitations for offset vector in word analogy question answering.

We propose to study any possible connection between offset similarity and word similarity, while framing the problem consistent with accepted definition of analogy; namely relational and structural mapping, where we map not only the entities, but also the relations.

Dataset	Accuracy	$\phi(x,y) < \phi(x,d)$	
		Count(True)	Count(False)
Sat	0.4613	114 (63.0%)	67 (37.0%)
Ekar	0.2912	116 (63.4%)	67 (36.6%)
U2	0.3833	91 (65.0%)	49 (35.0%)
Bats	0.7063	309 (58.5%)	219 (41.5%)
U4	0.3828	166 (62.4%)	100 (37.6%)
Total	-	**814 (61.5%)**	**510 (38.5%)**

Table 2: An analysis of incorrectly answered questions using the difference (offset) of word vectors across benchmarks, obtained from Fasttext. Given the query $x : x^*$, the answer $y : y^*$, and the distractor chosen by the offset vector as the answer $\boldsymbol{d_i} : \boldsymbol{d_i^*}$, let ψ be the cosine similarity and let $w : w^*$ be either the distractor or the true answer; then $\phi(x, w) = \max(\psi(x, w), \psi(x^*, w^*))$. The percentage in parentheses is relative to the total count of incorrectly answered questions. The last row reports the total counts (with corresponding percentages) across all datasets.

5 The Semantic Similarity and Offset Similarity

5.1 Evidence from Vector Offset Error Analysis

In word analogy question answering, when similarity between query and distractors exist, the offset vector may point to the the distractor most similar to the query as the answer, whereas when such similarity between query and the true answer exist, it helps with identifying the correct answer. Considering the true answer, and the incorrectly predicted answer by offset vector, it is often incorrectly chosen distractor. Accordingly, as can be seen in Table 2, when comparing the pairiwse similarity between query and true answer with that between query and incorrectly chosen distractor, the later is often greater than the former in incorrectly answered questions by offset vector across several benchmarks. To that end, using one sided binomial test of H_0, the statistical significance of the observation indicating the correspondence between pairwise similarity and offset similarity of the observation is shown in Section C.1, Table 7. As such, the plausible statement is that: offset vector tend to be similar to the words with which there is pairwise similarity.

5.2 Evidence from Various Permutations of Word Analogy

Based on the parallelogram model for word analogy, given a proportional word analogy $x : x^* :: y : y^*$, we can have further valid analogous permutations. For example, given King : Queen :: Man : Woman, we also have the proportional analogy King : Man :: Queen : Woman. As such, all permutations that result in an equal offset in the parallelogram (Figure 5) are valid proportional analogies, while those that violate offset equality are considered non-analogous and therefore invalid permutations. To that end, Prade and Richard (2013) meaningfully links various valid permutations with logic. Here, we use the permutation names as in their work. In addition, we consider two invalid permutations as well, namely $x : y^* :: y : x^*$ and $x : y^* :: x^* : y$ called here perpendicular and perpendicular_ respectively, as the corresponding vectors tend to be perpendicular in the parallelogram Figure 5. Note that the left-hand side in both perpendicular and perpendicular_ corresponds to the same vector, and the right-hand side corresponds to opposite vectors. An overview of various permutations is shown in Figure 5.

Assuming that the identity permutation is about the relation from one context to the relation in another context, such as color: colors:: computer: computers, the central permutation color: computer:: colors:: computers is the permutation wherein similarity due to relatedness and similarity due to semantics align: head words (color, colors) and tail words (computer, computers) each correspond to a unique context, whereas in the identity permutation, head words (color, computer) and tail words (colors, computers) are not from unique con-

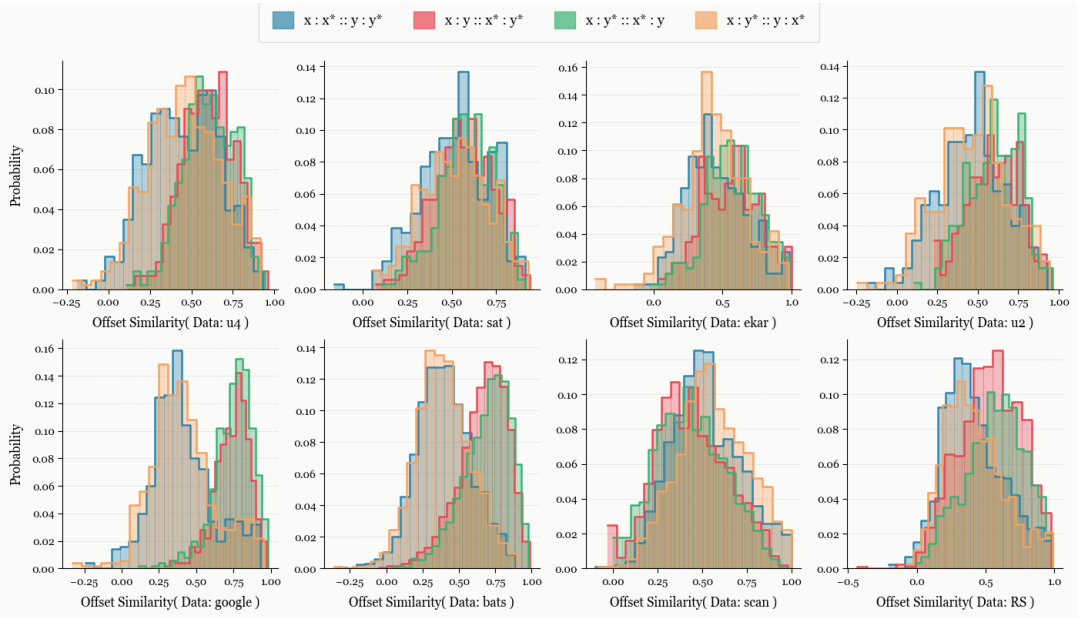


Figure 4: Distribution of offset similarity for various permutations of analogous terms with word embedding model Fasttext. As can be seen, the central and perpendicular permutations almost have similar distributions, whereas the identity and perpendicular_ permutations have similar distributions. Additionally, in some datasets, the central permutation is much closer to 1 than the identity permutation.

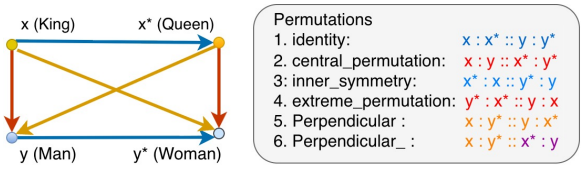


Figure 5: According to the parallelogram model for analogy, the offset vectors of analogous words for various valid permutations are equal. Here the 5th and 6th permutations are invalid, as the corresponding vectors are perpendicular.

texts. As such, in Figure 4, we can see that often the distribution of offset similarity for the central permutation is placed closer to 1 than the same for the identity permutation. Furthermore, among the invalid permutations perpendicular and perpendicular_, often offset similarity is greater for the perpendicular_ permutation. This in fact shows that when head and tail are from the same contexts, offset similarity is higher: $(x - y*, x * -y)$ both have the head words from the source context, while tail words are from the target context (similar to the central permutation). In the perpendicular permutation, $(x-y*, y-x*)$ it is the other way around (see Figure 4). According to our hypothesis stating that while constructing word analogy questions, when choosing distractors, preference is given to words with pairwise similarity with the query, we expect that with the offset vector, the inner_symmetry permutation $(x^* : x :: y^* : y)$ should be easier to solve as the direction of the offset reverses. In other words, if distractors (d_i, d_i^*) have pairwise head and tail similarity with word pair (x, x^*), then the query in the inner_symmetry permutation is less similar to distractors, as the head word in inner_symmetry is swapped with the tail word (x^*, x). Accordingly, to verify if the stated assumptions are indeed correct, according to the different permutations, in each word analogy question we replace the query and answer, and then, in addition to accuracy, we also calculate the average similarity in head and tail words between query, distractors, and the true answer. Additionally, in each case, the offset similarity is calculated as well. The result is shown in Table 3.

As such, Table 3 shows that the inner_sym permutation yields improved accuracy. The reason for this improvement is that distractors are chosen to be pairwise similar to the query under the identity permutation. As such, when head and tail are swapped in inner_symmetry, the direction of the relation changes. Additionally, the central permutation results in much improved performance, as the pairwise similarity between head and tail terms and the so-called relatedness similarity combine. As a takeaway from this experiment, we can state that, in word embedding models, related words are, to some extent similar. Moreover, in word analogy

Model	Permutation	w	sat			ekar			bats			google		
			avg(Q)	$\Phi(x,w)$	$\psi(r_x,r_w)$	avg(Q)	$\Phi(x,w)$	$\psi(r_x,r_w)$	avg(Q)	$\Phi(x,w)$	$\psi(r_x,r_w)$	avg(Q)	$\Phi(x,w)$	$\psi(r_x,r_w)$
Fasttext	identity	$w=y$	0.461	0.426	0.169	0.291	0.387	0.052	0.700	0.408	0.243	0.948	0.483	0.426
		$w=d_i$	0.539	0.459	0.087	0.709	0.478	0.026	0.300	0.536	0.012	0.052	0.630	0.004
	inner_sym	$w=y$	0.771	0.426	0.169	0.391	0.387	0.052	0.740	0.408	0.243	0.966	0.483	0.426
		$w=d_i$	0.229	0.459	-0.087	0.609	0.478	-0.026	0.260	0.536	-0.012	0.034	0.630	-0.004
	central	$w=y$	0.807	0.469	0.230	0.636	0.482	0.190	0.930	0.602	0.473	0.998	0.711	0.677
		$w=d_i$	0.193	0.434	-0.002	0.364	0.430	0.016	0.070	0.432	-0.001	0.002	0.509	0.001
	extreme	$w=y$	0.768	0.469	0.230	0.705	0.481	0.189	0.930	0.602	0.473	0.998	0.711	0.677
		$w=d_i$	0.232	0.438	-0.001	0.295	0.424	-0.021	0.070	0.443	-0.000	0.002	0.522	-0.002
Word2vec	identity	$w=y$	0.410	0.504	0.167	0.110	0.506	0.070	0.650	0.533	0.215	0.700	0.635	0.358
		$w=d_i$	0.590	0.556	0.081	0.890	0.632	0.022	0.350	0.638	0.008	0.300	0.761	0.023
	inner_sym	$w=y$	0.730	0.504	0.167	0.180	0.506	0.070	0.690	0.533	0.215	0.730	0.635	0.358
		$w=d_i$	0.270	0.556	-0.081	0.820	0.632	-0.022	0.310	0.638	-0.008	0.270	0.761	-0.023
	central	$w=y$	0.750	0.548	0.247	0.230	0.616	0.274	0.920	0.678	0.479	0.800	0.840	0.707
		$w=d_i$	0.250	0.517	0.013	0.770	0.570	0.009	0.080	0.575	-0.001	0.200	0.675	0.003
	extreme	$w=y$	0.760	0.548	0.247	0.260	0.616	0.274	0.930	0.678	0.479	0.730	0.840	0.707
		$w=d_i$	0.240	0.567	-0.003	0.740	0.568	-0.006	0.070	0.555	-0.000	0.270	0.634	-0.000

Table 3: Accuracy and Similarity across various permutations of word analogies in different benchmarks. ψ is cosine similarity. avg(Q) stands for average questions for which relation similarity of the query($x : x^*::\underline{?}$) is considered most similar to $w = y$ by the corresponding model, which is the total question correctly answered (accuracy). For $w \in [y, d_i]$, w stands for word pair (w, w^*). r_w is the offset corresponding to (w, w^*). $\Phi(x, w_i) = \frac{1}{n}\sum_{i=1}^{n} \frac{\psi(x,w_i)+\psi(x^*,w_i^*)}{2}$. Note: when $w = y$, $n = 1$, and when $w = d_i$ n is the number of choices in the question and is shown in Table 1. Distractors (d_i) are non-answer choices. For all datasets and models, if the accuracy for inner_sym and central permutations is greater than that of identity permutation, they are highlighted.

Model	Permutation	sat	ekar	u2	u4
gpt-4.1-mini	identity	0.579	0.538	0.801	0.695
	inner_sym	0.609	0.590	0.808	0.668
	central	0.584	0.779	0.830	0.737
	extreme	0.667	0.764	0.896	0.779
gpt-5.2	identity	0.705	0.766	0.947	0.807
	inner_sym	0.720	0.799	0.950	0.835
	central	0.770	0.951	0.988	0.864
	extreme	0.978	0.954	0.990	0.859

Table 4: Comparative performance of gpt-4.1-mini and gpt5.2 across permutations.

benchmarks, to an extent, distractors are chosen to be semantically similar to the query, however not relationally similar to the query or the true answer. Furthermore, word embedding model may capture the direction of relations through offset vector.

We conducted a similar experiment with LLMs. Using ChatGPT API calls, we asked question constructed according to various permutations from models: gpt-5.2 and gpt-4.1-mini as can be seen in the Table 4. To that end, the mentioned models too often exhibit improved performance on inner_sym and central permutations than the identity permutations.

5.3 Evidence From Word Analogy Questions: Google

Using the columnized version of the word analogy dataset Google [1] (Mikolov et al., 2013b), we show that what makes solving word analogy questions with offset vectors difficult is the semantic similarity between the query and distractors. As such, given that the dataset Google covers 14 relation types, such as, 1. capital-common-countries (e.g., Athens, Greece) 2. currency (e.g., USA dollar), we first sort word pairs into relation categories, and then from each given relation, sample two pair of words as the query and the answer respectively. To that end, we make two versions of word analogy questions, Easy and Hard. In the Hard version, distractors are chosen such that given the query (x, x^*), and a distractor d_i, d_j^*, there are word pairs such that (d_i, d_i^*), and (d_j, d_j^*) are analogous to the query (We also make sure neither $d_i = x$ nor $d_i^* = x^*$). This makes the distractors of the Hard variant semantically similar to the query, as shown in Figure 1; however, in the Easy version, distractors are sampled from a relation category other than the query's. As shown in Tables: 5 and 6, across various word embedding models, performances are often lower on the Hard version when distractors

[1] https://huggingface.co/datasets/almogtavor/google-analogy-dataset

are semantically similar to the query. Moreover, in the Appendix, Section C.1 Table 9 the Mcnemar's exact test show that the performance e difference on Easy and Hard questions are significant . Moreover, in the so called Easy questions, answering word analogy question using the similarity between head and tail words (x, y), and $(x*, y*)$ respectively, can result in better or equivalent performance as offset similarity (Table 5). However, when distractors (d_i, d_i^*) are chosen to have pairwise head and tail similarity with the query, as in Hard questions, offset similarity achieves relatively better result or performance is close to random.

Category	Model	Easy	Hard	Diff
MLM	BERT-Large*	**92.7**	**65.0**	27.7
	BERT-Large	88.0	29.0	59.0
	RoBERTa-Large*	**71.6**	**55.0**	16.6
	RoBERTa-Large	62.0	19.0	43.0
Causal	GPT-2*	45.9	**25.6**	20.3
	GPT-2	**62.0**	21.6	40.4
Enc–Dec	T5-Large*	24.7	16.0	8.7
	T5-Large	**29.7**	**18.0**	11.7
Other	FastText*	**84.0**	**47.0**	37.0
	FastText	71.0	17.0	54.0

Table 5: Multi-choice analogy questions (237 questions in each category) answering. For each model, * indicates offset similarity was used; otherwise pairwise similarity between head and tail terms was used. Diff denotes: Easy- Hard settings. Bold indicates the better accuracy between the offset (*) and pairwise variants for each model, per column. A statistical significance test is shown in the Appendix, Section C.1, Table 9.

6 Conclusion

In this paper, we show that related words are to an extent similar. Additionally, word embedding model capture the direction of relation between words evidenced by performance gap between inner_sym permutation and identity with offset vector. Furthermore, the pairwise similarity influences offset similarity between words, and hence can be viewed as limitation for offset vector as pairwise similarity and relation similarity can be different. As such, the offset of words (Paris-Belgium) ideally should not equal the offset of words (Berlin-Germany).

Relation (r)	Easy	Hard	Diff
capitalcountry	0.763	0.342	0.421
countrycurrency	1.000	0.000	1.000
citystate	0.625	0.219	0.406
malefemale	0.889	0.222	0.667
adjadverb	0.667	0.250	0.417
adjantonym	0.667	0.333	0.333
adjcomparative	0.941	0.471	0.471
adjsuperlative	1.000	0.800	0.200
verbparticiple	0.944	0.667	0.278
countrynationality	0.967	0.533	0.433
verbpreterite	1.000	0.609	0.391
nounplural	1.000	0.875	0.125
verbplural	1.000	0.625	0.375

Table 6: Performance for offset vectors in the word embedding model FastText on different relations. Diff denotes: Easy - Hard. As shown, in FastText, some relations, such as countrycurrency, are particularly vulnerable to pairwise similarity between the query and distractors.

Limitations

This paper relies on empirical evidences, and the theoretical background for the observations is not explored.

Ethical Considerations

This work is in coherence with ACL code of conduct and ethical considerations.

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A Reproducibility

The code to reproduce the results can be accessed in the following links: https://github.com/bahramiramazan/offset_and_semantic

B Word Embedding Models Considered

Various word embedding models considered in this study are 1. fastext (Bojanowski et al., 2017) 2. bert-large-uncased (Devlin et al., 2018) 3. roberta-large (Liu et al., 2019) 4. gpt2 (Cheng et al., 2024) 5. T5-large (Raffel et al., 2020) 6. word2vec (Mikolov et al., 2013a)

C Experimental Setup

For GPT models considered in this experiments, gpt-4.1-mini, and gpt-5.2 respectively, using the api, we prompted the model to answer the multiple choice x is to x^* as : 1. y is to y^* 2. d_i is to d_j^* 3. ..etc. The GPT models were accessed during 01.02.2026 to 15.07.2026. For word embedding models: Fasttext, and word2vec, we simply obtained the word embeddings. For other neural based language models: berta-large-uncased, roberta-large, t5-large, and gpt2, we first feed the word pair sentence as as x is to x^* into the model, and then slice the embedding of words from the corresponding sentence embeddings.

C.1 Significance Test

For various observations in this paper, we also conduct appropriate statistical tests to assess their significance. As such, to show that pairwise similarity between head and tail words are not a random observation in all incorrectly answered questions by offset vector in the word embedding model Fasttext, we use a one-sided binomial test with H_0 as is shown in the Table 7. Additionally, to compare performance across pairs of permutations, we use McNemar's exact test; as for given questions from two given permutations, we can treat performance on given permutations as a paired observation. As such, in each pair, the query and answer differ based on the permutations, while the distractors stay the same in both versions of the questions corresponding to each of the permutations. The test result is shown in Table 8. Similarly, for comparing model performance on Google_Easy and Google_Hard each question has two versions . Note that the query and answer are the same in both versions, while distractors change. As such we here we McNemar's exact test as well. The result is shown in the Table 9.

Dataset	n	**Count(True)**	$\hat{p}$	z	p**-value**
Sat	181	114	0.630	3.49	2.4×10^{-4}
Ekar	183	116	0.634	3.62	1.5×10^{-4}
U2	140	91	0.650	3.55	1.9×10^{-4}
Bats	528	309	0.585	3.92	4.5×10^{-5}
U4	266	166	0.624	4.05	2.6×10^{-5}
Total	**1324**	**814**	**0.615**	**8.35**	$< \mathbf{10^{-15}}$

Table 7: One-sided binomial test of Null Hypothesis $H_0 : p = 0.5$ against $H_1 : p > 0.5$, where p is the probability that $\phi(x, y) < \phi(x, d)$ holds among incorrectly answered questions. Note: $\hat{p}$ is the observed proportion of True cases, and z is the normal-approximation statistic (without continuity correction). 1. † Exact binomial one-sided p-value used because n is small (normal approx. yields $p = 0.025$). 2. All datasets are individually significant at $\alpha = 0.05$.

Model	Dataset	Pair Comparison	Accuracies (p_1 vs p_2)	b / c	p-value	Holm-adj. p	Sig.
Fasttext	SAT ($N = 336$)	identity vs. inner_sym	0.46 vs 0.77	25/129	< 0.0001	< 0.0001	Yes
		identity vs. central	0.46 vs 0.81	20/136	< 0.0001	< 0.0001	Yes
		identity vs. extreme	0.46 vs 0.77	21/124	< 0.0001	< 0.0001	Yes
		central vs. extreme	0.81 vs 0.77	40/27	0.1421	0.5388	No
	EKAR ($N = 261$)	identity vs. inner_sym	0.29 vs 0.39	26/52	0.0043	0.0303	Yes
		identity vs. central	0.29 vs 0.64	15/105	< 0.0001	< 0.0001	Yes
		identity vs. extreme	0.29 vs 0.70	14/122	< 0.0001	< 0.0001	Yes
		central vs. extreme	0.64 vs 0.70	21/39	0.0273	0.1640	No
	U2 ($N = 227$)	identity vs. inner_sym	0.38 vs 0.66	14/76	< 0.0001	< 0.0001	Yes
		identity vs. central	0.38 vs 0.88	3/115	< 0.0001	< 0.0001	Yes
		identity vs. extreme	0.38 vs 0.87	3/113	< 0.0001	< 0.0001	Yes
		central vs. extreme	0.88 vs 0.87	16/14	0.8555	1.0000	No
	U4 ($N = 431$)	identity vs. inner_sym	0.38 vs 0.61	37/134	< 0.0001	< 0.0001	Yes
		identity vs. central	0.38 vs 0.87	9/219	< 0.0001	< 0.0001	Yes
		identity vs. extreme	0.38 vs 0.90	5/227	< 0.0001	< 0.0001	Yes
		central vs. extreme	0.87 vs 0.90	19/31	0.1189	0.5388	No
	Google ($N = 499$)	identity vs. inner_sym	0.95 vs 0.97	8/17	0.1078	0.5388	No
		identity vs. central	0.95 vs 1.00	1/26	< 0.0001	< 0.0001	Yes
		identity vs. extreme	0.95 vs 1.00	1/26	< 0.0001	< 0.0001	Yes
		central vs. extreme	1.00 vs 1.00	1/1	1.0000	1.0000	No

Table 8: McNemar's exact test comparing permutation performance of Fasttext on the same questions. b (c): questions answered correctly only under the first (second) permutation. p-values are two-sided; Holm correction is applied across all 20 comparisons. Significance at $\alpha = 0.05$ after correction.

	Easy vs. Hard				Easy* vs. Hard*			
Model	b	c	p-value	Sig.	b	c	p-value	Sig.
RoBERTa-Large	109	14	2.08×10^{-19}	***	65	29	2.62×10^{-4}	***
BERT-Large	142	11	3.54×10^{-30}	***	73	12	8.09×10^{-12}	***
T5-Large	55	29	6.00×10^{-3}	**	48	29	3.95×10^{-2}	*
GPT-2	100	9	1.44×10^{-20}	***	71	26	5.46×10^{-6}	***
FastText	128	7	6.72×10^{-30}	***	94	7	1.46×10^{-20}	***

Table 9: McNemar's exact test results comparing model performance (Easy vs. Hard and Easy* vs. Hard*).For each model, * indicates offset similarity was used; otherwise pairwise similarity between head and tail terms was used. Here, b and c denote the discordant-pair counts. Significance levels: $^{*}\, p < 0.05$, $^{**}\, p < 0.01$, $^{***}\, p < 0.001$.